

# Robustness of Classification Models under Saliency-Guided Occlusion or Removal: A Cross-Modal Empirical Analysis

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## Abstract

Classification models deployed in real-world settings frequently encounter inputs where decision-relevant features are missing or occluded. Existing robustness evaluations and analysis focus on generic corruptions and distribution shifts, without considering the removal of features that the model itself identifies as important. We present a systematic empirical analysis of model robustness under saliency-guided feature removal across two modalities — image and tabular data. Using SHAP and LIME to identify decision-relevant features, we progressively remove them and measure predictive degradation via a normalized cross-entropy-based robustness score. We analyse ImageNet, ImageNet-C, ImageNet-R, and the OpenML-CC18 suite across ResNet-50, MLP, Transformer, XGBoost, and Random Forest, finding that robustness varies substantially across architectures and modalities, providing empirical guidance for architecture selection.

## Keywords

Empirical Analysis, Classification, Robustness, Explainable Artificial Intelligence, Saliency Maps, Data Augmentation

## 1 Introduction

In real-world classification scenarios, data is often imperfect. Objects may be partially missing, hidden, or overlapped by other elements in the scene. Ensuring that classifiers remain reliable under such conditions is essential, particularly in safety-critical or high-stakes applications [8, 28]. This motivates a focused investigation into how models behave when decision-relevant parts of the data become unavailable.

It is insufficiently understood how different model architectures respond to the systematic occlusion or removal of decision-relevant features. This gap is further compounded by the lack of cross-modal analyses, leaving it unknown whether robustness behavior under feature removal is architecture-specific, modality-specific, or a more general phenomenon.

Without a systematic empirical analysis, practitioners lack guidance on which architectures are inherently more robust to feature occlusion or removal. Understanding how and to what extent different model architectures degrade under feature removal is therefore a prerequisite for informed model selection and deployment.

Existing robustness evaluations predominantly assess model behavior under generic corruptions such as noise, blur, or distribution shifts [8, 13, 28]. Established augmentation strategies are designed and evaluated within this same perturbation space [7, 14, 29]. These evaluations do not address the removal of features the model itself identifies as decision-relevant. Benchmark datasets are mainly confined to a specific domain, and comparative analyses typically remain within a single modality [9, 12, 13].

Our contribution is an empirical analysis that systematically evaluates the robustness of classification models across two modalities — image and tabular data — under saliency-guided

feature removal. The analysis covers convolutional networks, Transformers, and tree-based models, spanning both neural and non-neural paradigms. For evaluation and comparison, we apply Explainable Artificial Intelligence (XAI) methods to identify decision-relevant features and progressively remove them. We compute the resulting predictive degradation by measuring the cross-entropy between the original prediction and the prediction after feature removal. This degradation across removal intensities is then aggregated to obtain a robustness score.

Our experiments reveal that robustness under occlusion or removal varies across architectures and data modalities. We analyze ImageNet [6], ImageNet-C [13], and ImageNet-R [12] for image data, and the OpenML-CC18 [2] suite for tabular data, observing divergent degradation behavior across ResNet-50 [11], MLP, Transformer [27], XGBoost [4], and Random Forest [3]. The findings provide empirical guidance for architecture selection in real-world applications.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 introduces the robustness metric and methodology. Section 4 describes the experimental setup. Sections 5 and 6 present and discuss the results.

## 2 Related Work

This section reviews related work that forms the foundation for the comparison metric used in the empirical analysis.

### 2.1 Explainable Artificial Intelligence

There are many XAI methods post-hoc analysis on how a model is working [5, 17, 20, 21, 23, 24]. Often XAI methods attribute model predictions to input features by computing saliency maps. Saliency maps are numerical scores indicating feature relevance for a given prediction. Many of them are modality-agnostic [22]. For instance, SHAP [17] and LIME [21] apply to both image and tabular data. SHAP computes feature attributions based on cooperative game theory and generalizes across neural and non-neural architectures. LIME generates local linear approximations of model behavior by perturbing inputs and fitting interpretable models to prediction changes.

### 2.2 Data Augmentation

A common strategy for improving model robustness is data augmentation, which exposes models to perturbed training inputs [19]. Popular robustness-oriented augmentation methods are CutMix [29] and AugMix [14]. CutMix replaces image regions with content from other samples and adjusts labels proportionally. AugMix constructs stochastic mixtures of corruption operations and enforces prediction consistency between clean and augmented inputs.

A related line of work integrates saliency information into augmentation pipelines [1, 10, 25, 26], using saliency maps to guide masking, mixing, or region selection during training. Although these approaches incorporate model-derived relevance

information, their objective is to improve predictive performance rather than to quantify robustness. Furthermore, the effect of saliency-guided manipulations is typically evaluated via standard accuracy metrics rather than robustness measures [1, 10, 25, 26].

### 2.3 Robustness Evaluation Benchmarks

Robustness is commonly assessed using standardized benchmark datasets that introduce controlled corruptions or domain shifts. For example, ImageNet-C [13] evaluates performance under algorithmically generated corruptions such as noise, blur, and weather effects. ImageNet-R [12] measures generalization to stylized renditions of natural images.

These benchmarks quantify performance degradation under predefined perturbation distributions but do not consider model-specific relevance information. Furthermore, robustness analyses are typically restricted to a single modality.

### 2.4 Summary and Positioning

The surveyed literature treats XAI, data augmentation for robustness, saliency-guided data augmentation, and robustness benchmarking as mainly separate areas. Robustness benchmarks rely on predefined perturbation distributions, while XAI methods are primarily used for explanation rather than quantitative robustness assessment. Optimization methods are either not using saliency maps or optimizing overall model performance and only optimizing robustness indirectly.

In contrast, our approach integrates these perspectives via a metric that uses saliency maps derived from XAI methods to define model-specific perturbations and measure robustness via the cross-entropy between the original prediction and the prediction after feature removal.

## 3 Robustness Evaluation Metric Calculation

This section describes how the metric for the comparison is calculated.

### 3.1 Notation

A dataset  $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$  of  $N$  datapoints is considered, where  $\mathbf{x}_i \in \mathbb{R}^d$  denotes the input features and  $y_i \in \{1, \dots, C\}$  the class label. For tabular data,  $d$  is the number of input features. For image data,  $d = H \times W \times Ch$ . A classification model  $f_\theta : \mathbb{R}^d \rightarrow \mathbb{R}^C$  is trained on the dataset  $\mathcal{D}$ . The predicted probability distribution for each datapoint is  $\mathbf{p}_i = \text{softmax}(f_\theta(\mathbf{x}_i)) \in \mathbb{R}^C$ .

### 3.2 Saliency Map Generation

Let  $\mathcal{S} : \mathbb{R}^d \times f_\theta \rightarrow \mathbb{R}^d$  denote an XAI method applied to the trained model  $f_\theta$ . For each correctly classified datapoint  $\mathbf{x}_i$ , a saliency map  $\mathbf{s}_i = \mathcal{S}(\mathbf{x}_i, f_\theta) \in \mathbb{R}^d$  is computed. Only correctly classified samples are here and in the following considered, as perturbing an already incorrect prediction does not yield interpretable degradation information.

### 3.3 Synthetic Sample Generation

For each datapoint  $\mathbf{x}_i$  and intensity level  $k \in \{k_1, \dots, k_m\}$ , a synthetic sample  $\mathbf{x}_i^{(k)}$  is generated by removing the top- $k$  percentage of most relevant pixels according to the saliency map  $\mathbf{s}_i$ .

Some XAI methods produce relevance outputs in the form of aggregated or grouped measurements. In such cases, an individual strategy is required to determine how the removal of individual pixels in a group is performed during the generation

of synthetic samples. Strategy used in this empirical analysis is stated in Section 4.3.

### 3.4 Quantifying Prediction Degradation

For each synthetic sample  $\mathbf{x}_i^{(k)}$ , the model’s predicted probability distribution  $\mathbf{p}_i^{(k)}$  is computed. To quantify the degradation in prediction quality, the cross-entropy  $\mathcal{L}_i^{(k)} = H(\mathbf{p}_i, \mathbf{p}_i^{(k)})$  between the original prediction distribution  $\mathbf{p}_i$  and each perturbed prediction distribution  $\mathbf{p}_i^{(k)}$  is calculated [18].

### 3.5 Robustness Metric Computation

The cross-entropy losses are aggregated over all datapoints and intensity levels and the result is normalized by the total number of losses with

$$R = \frac{1}{nN} \sum_{k=1}^n \sum_{i=1}^N \mathcal{L}_i^{(k)}. \quad (1)$$

A low value of  $R$  indicates that  $\mathbf{p}_i^{(k)} \approx \mathbf{p}_i$  across removal intensities, reflecting distributional stability of the model’s predictions under feature removal. A high value of  $R$  indicates substantial divergence between  $\mathbf{p}_i^{(k)}$  and  $\mathbf{p}_i$ .

## 4 Experimental Apparatus

This section describes the setup for the empirical analysis.

### 4.1 Datasets

Table 1 provides an overview of the image datasets and their basic statistics. ImageNet is used as a benchmark for natural image classification [6]. ImageNet-C is included to study robustness under common, systematically controlled corruptions — e.g., noise and blur — while keeping the underlying task unchanged [13]. ImageNet-R complements this by targeting a natural domain shift, thereby testing whether observations persist under distribution changes beyond synthetic corruptions [12].

For tabular data, the OpenML-CC18 benchmarking suite [2] is used. It is a collection of 72 classification tasks drawn from the OpenML platform. The suite covers a diverse range of domains, feature types, and class structures, with datasets ranging from 500 to 100,000 observations and up to 5,000 features.

### 4.2 Classification Models

An overview of all model–dataset combinations is given in Table 2. For image data, ResNet-50 [11], MLP, and Transformer [27] are employed. Together, these three models cover convolutional, fully-connected, and self-attention paradigms, providing a representative cross-section of widely used neural architectures. For tabular datasets, XGBoost [4] and Random Forest [3] are used. The model selection is motivated by their prevalence in the literature and their coverage of distinct inductive biases, enabling a comparative robustness analysis across fundamentally different model families.

### 4.3 Parameters of Metric Computation

The robustness metric introduced in Section 3 is computed using the following parameters.

SHAP [17] and LIME [21] are employed to calculate the saliency maps. SHAP computes Shapley values that are defined over the same input space as the data and are used directly as saliency maps. LIME for image data partitions the input into contiguous

regions of semantically coherent pixels, referred to as superpixels. Relevance scores are assigned at the superpixel level and not at the individual pixel level. When a removal intensity threshold falls within a superpixel — meaning only a fraction of that superpixel’s pixels need to be removed to reach the target removal percentage  $k$  — individual pixels are sampled uniformly at random from that superpixel to fill the remaining quota.

At each intensity level,  $k \in \{4, 8, 16, 32\}$  % of the most decision-relevant features per sample are removed. These levels provide coverage from minimal to substantial removal while staying below the regime where predictions become uninformative due to excessive information loss. For image data, pixels to remove are replaced with the median value of the removed pixels, providing a neutral substitute to avoid out-of-distribution artifacts. For tabular data, removed values are set to zero.

#### 4.4 Experimental Procedure

All models are trained independently on each dataset of their respective modality using the hyperparameters determined in Appendix D. Predefined dataset splits are not used. Instead, stratified cross-validation is applied uniformly across all datasets to ensure class-balanced evaluation folds and reduce the influence of dataset-specific partitions on reported results [15]. Model weights and output logits for all samples are saved after training for use in the subsequent metric computation.

Using the parameters defined in Section 4.3, the robustness metric  $R$  is computed for each combination of model, dataset, and XAI method as described in Section 3.

## 5 Results

## 6 Discussion

## 7 Conclusion

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## A Use of Generative AI

During the preparation of this work, the authors used generative AI tools (OpenAI ChatGPT, Anthropic Claude) for proofreading and rephrasing of text passages. All AI-generated suggestions were reviewed, verified, and edited by the authors. The authors take full responsibility for the content of this publication.

## B Datasets

| Dataset    | Setting      | #Classes | #Samples   | Citations |
|------------|--------------|----------|------------|-----------|
| ImageNet   | –            | 80,841   | 50,000,000 | [6]       |
| ImageNet-C | Corruption   | 1,000    | 50,000     | [13]      |
| ImageNet-R | Domain shift | 200      | 30,000     | [12]      |

**Table 1: Image datasets used in this work.**

## C Model and Dataset Combinations

| Data Modality | Models                                | Datasets/ Suits                                |
|---------------|---------------------------------------|------------------------------------------------|
| Image         | ResNet-50 [11], MLP, Transformer [27] | ImageNet [6], ImageNet-C [13], ImageNet-R [12] |
| Tabular       | XGBoost [4], Random Forest [3]        | OpenML-CC18 [2]                                |

**Table 2: Overview of classification models and datasets.**

## D Hyperparameter

Hyperparameters for all models are optimized via grid search with stratified cross-validation on the training set [15, 16]. The optimization objective is classification accuracy for all models. Search spaces and selected configurations are designed to balance computational feasibility with thorough exploration of parameter ranges.

For ResNet-50 [11], we tune learning rate, batch size, and weight decay during fine-tuning. For MLP, we tune the number of hidden layers, layer width, learning rate, and dropout rate. For Transformer [27], we tune the number of attention heads, number of encoder layers, model dimension, learning rate, and dropout rate. For XGBoost [4] and Random Forest [3], we optimize tree depth, number of estimators, and regularization parameters.

For all random operations (e.g., shuffling and subsampling) a fixed random seed of 42 is used to ensure reproducibility across runs.